Early Sepsis Prediction from ICU Time Series: Does Modelling Missingness Patterns Improve Prediction?

Project Synopsis

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1 Motivation and Problem Statement

Sepsis is a leading cause of intensive care unit (ICU) mortality, and delays in treatment are strongly associated with worse patient outcomes. Early sepsis, however, presents with subtle, easily overlooked physiological changes that closely resemble normal clinical variation, making early detection difficult even for experienced clinicians. Automated early-warning systems built on ICU time-series data offer a way to flag at-risk patients hours before clinical recognition, potentially enabling earlier intervention.

A central challenge in this setting is that ICU data exhibits severe and *non-random* missingness: laboratory tests are ordered selectively based on clinical suspicion rather than at fixed intervals, so the presence or absence of a measurement is itself informative. Most existing approaches treat missing data purely as an obstacle to be imputed away, discarding this signal. This project investigates whether explicitly modelling missingness patterns, rather than merely filling them in, improves early sepsis prediction.

2 Related Work and Positioning

Lipton et al. [1] demonstrated that treating missingness indicators as explicit features, rather than relying on imputation alone, improves predictive performance in clinical time-series classification. Che et al. [2] extended this idea with the GRU-D architecture, formally identifying *informative missingness* as an underexploited signal in multivariate clinical time series. Both works, however, rely on deep recurrent architectures whose internal handling of missingness is difficult to interpret in isolation.

Within the PhysioNet/CinC 2019 Sepsis Challenge specifically, the majority of published approaches use gradient-boosted tree ensembles (e.g., XGBoost) with imputation strategies such as mean-filling, linear interpolation, or MissForest [4]. Where missingness indicators are used, they are typically bundled together with many other engineered features without isolating their individual contribution to performance.

This project differs by running a **controlled comparison** on the same classifiers across three explicit data representations: (A) standard imputation only, (B) imputation plus binary missingness masks, and (C) imputation plus time-since-last-measurement features. Holding the model architecture fixed while varying only the missingness representation isolates the specific contribution of missingness modelling to prediction quality, a comparison that is largely absent from prior work on this dataset.

3 Dataset

We use the PhysioNet/Computing in Cardiology Challenge 2019 dataset [3], comprising hourly ICU records for 40,336 patients across two hospital systems (Hospital A: 20,336 patients; Hospital B: 20,000

patients), totaling 1,552,210 patient-hours. Each hourly record contains 8 vital signs, 26 laboratory values, and 6 demographic variables, with a binary `SepsisLabel` marking sepsis onset per Sepsis-3 criteria, shifted six hours earlier than clinical recognition to reward early prediction.

Exploratory analysis of the full dataset confirms the challenges motivating this project. Class imbalance is severe at the row level: 7.27% of patients ever develop sepsis, but only 1.80% of individual hourly rows are labeled positive. Missingness is pronounced and highly uneven across features, as summarized in Table 1.

Table 1: Missingness in the five worst-affected laboratory features.

Feature	**% Missing**
Bilirubin_direct	99.81
Fibrinogen	99.34
TroponinI	99.05
Bilirubin_total	98.51
Alkalinephos	98.39

This level of missingness is consistent with selective, clinically-driven test ordering rather than random sensor dropout, making the dataset directly suited to studying informative missingness. Its open access, established provenance, and manageable scale make it feasible to work with within the course timeframe.

4 Performance Metrics

Given the severe row-level class imbalance (98.2% negative), standard accuracy is uninformative: a trivial classifier that always predicts "no sepsis" would score 98.2%. We therefore report:

- **PR-AUC** (precision–recall area under curve), which is more sensitive to minority-class performance than ROC-AUC under severe imbalance;
- **F1-score** at a selected operating threshold, balancing precision and recall;
- The Challenge's own **time-dependent utility score**, which rewards predictions made between 12 hours before and 3 hours after true sepsis onset, and penalizes both late detection and false alarms.

The utility score is essential to this project because it captures the *clinical* cost structure of an early-warning system in a way that PR-AUC and F1 do not: a model can achieve strong discriminative performance while still alerting too late to be clinically useful. Comparing representations A, B, and C across all three metrics will indicate whether missingness-aware modelling improves not only raw discrimination, but the clinically-relevant early-detection property the utility score is designed to capture.

References

[1] Z. C. Lipton, D. C. Kale, and R. Wetzel, "Modeling Missing Data in Clinical Time Series with RNNs," *arXiv preprint arXiv:1606.04130*, 2016.

[2] Z. Che, S. Purushotham, K. Cho, D. Sontag, and Y. Liu, "Recurrent Neural Networks for Multivariate Time Series with Missing Values," *Scientific Reports*, vol. 8, no. 6085, 2018.

[3] M. A. Reyna, C. S. Josef, R. Jeter, S. P. Shashikumar, M. B. Westover, S. Nemati, A. Sharma, and G. D. Clifford, "Early Prediction of Sepsis From Clinical Data: The PhysioNet/Computing in Cardiology Challenge," *Critical Care Medicine*, vol. 48, no. 2, pp. 210–217, 2019.

[4] Survey of PhysioNet/CinC 2019 Challenge entries (e.g., XGBoost-based pipelines using mean imputation, linear interpolation, and MissForest for missing-value handling), as reported in Challenge proceedings papers, Computing in Cardiology, 2019.